

OpenStax Calculus Volume 3

Section 5.7 Exercises: Change of Variables in Multiple Integrals

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In the following exercises, the function $T : S \rightarrow R, T(u, v) = (x, y)$ on the region $S = \{(u, v) | 0 \leq u \leq 1, 0 \leq v \leq 1\}$ bounded by the unit square is given, where $R \subset \mathbb{R}^2$ is the image of S under T .

- (a) Justify that the function T is a C^1 transformation.
- (b) Find the images of the vertices of the unit square S through the function T .
- (c) Determine the image R of the unit square S and graph it.

1. $x = 2u, y = 3v$
2. $x = \frac{u}{2}, y = \frac{v}{3}$
3. $x = u - v, y = u + v$
4. $x = 2u - v, y = u + 2v$
5. $x = u^2, y = v^2$
6. $x = u^3, y = v^3$

In the following exercises, determine whether the transformations $T : S \rightarrow R$ are one-to-one or not.

7. $x = u^2, y = v^2$, where S is the rectangle of vertices $(-1, 0), (1, 0), (1, 1)$, and $(-1, 1)$.
8. $x = u^4, y = u^2 + v$, where S is the triangle of vertices $(-2, 0), (2, 0)$, and $(0, 2)$.
9. $x = 2u, y = 3v$, where S is the square of vertices $(-1, 1), (-1, -1), (1, -1)$, and $(1, 1)$.
10. $T(u, v) = (2u - v, u)$, where S is the triangle of vertices $(-1, 1), (-1, -1)$, and $(1, -1)$.
11. $x = u + v + w, y = u + v, z = w$, where $S = R = \mathbb{R}^3$.
12. $x = u^2 + v + w, y = u^2 + v, z = w$, where $S = R = \mathbb{R}^3$.

In the following exercises, the transformations $T : S \rightarrow R$ are one-to-one. Find their related inverse transformations $T^{-1} : R \rightarrow S$.

13. $x = 4u, y = 5v$, where $S = R = \mathbb{R}^2$.
14. $x = u + 2v, y = -u + v$, where $S = R = \mathbb{R}^2$.
15. $x = e^{2u+v}, y = e^{u-v}$, where $S = \mathbb{R}^2$ and $R = \{(x, y) | x > 0, y > 0\}$

16. $x = \ln u, y = \ln(uv)$, where $S = \{(u, v) | u > 0, v > 0\}$ and $R = \mathbb{R}^2$.
17. $x = u + v + w, y = 3v, z = 2w$, where $S = R = \mathbb{R}^3$.
18. $x = u + v, y = v + w, z = u + w$, where $S = R = \mathbb{R}^3$.

In the following exercises, the transformation $T : S \rightarrow R, T(u, v) = (x, y)$ and the region $R \subset \mathbb{R}^2$ are given. Find the region $S \subset \mathbb{R}^2$.

19. $x = au, y = bv, R = \{(x, y) | x^2 + y^2 \leq a^2b^2\}$, where $a, b > 0$
20. $x = au, y = bv, R = \{(x, y) | \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\}$, where $a, b > 0$
21. $x = \frac{u}{a}, y = \frac{v}{b}, z = \frac{w}{c}, R = \{(x, y, z) | x^2 + y^2 + z^2 \leq 1\}$, where $a, b, c > 0$
22. $x = au, y = bv, z = cw, R = \{(x, y, z) | \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} \leq 1, z > 0\}$, where $a, b, c > 0$

In the following exercises, find the Jacobian J of the transformation.

23. $x = u + 2v, y = -u + v$
24. $x = \frac{u^3}{2}, y = \frac{v}{u^2}$
25. $x = e^{2u-v}, y = e^{u+v}$
26. $x = ue^v, y = e^{-v}$
27. $x = u \cos(e^v), y = u \sin(e^v)$
28. $x = v \sin(u^2), y = v \cos(u^2)$
29. $x = u \cosh v, y = u \sinh v, z = w$
30. $x = v \cosh(\frac{1}{u}), y = v \sinh(\frac{1}{u}), z = u + w^2$
31. $x = u + v, y = v + w, z = u$
32. $x = u - v, y = u + v, z = u + v + w$

33. The triangular region R with the vertices $(0, 0)$, $(1, 1)$, and $(1, 2)$ is shown in the following figure.
[Figure: A triangle with corners at the origin, $(1, 1)$, and $(1, 2)$.]

- (a) Find a transformation $T : S \rightarrow R, T(u, v) = (x, y) = (au + bv, cu + dv)$, where a, b, c , and d are real numbers with $ad - bc \neq 0$ such that $T^{-1}(0, 0) = (0, 0)$, $T^{-1}(1, 1) = (1, 0)$, and $T^{-1}(1, 2) = (0, 1)$.
- (b) Use the transformation T to find the area $A(R)$ of the region R .

34. The triangular region R with the vertices $(0, 0)$, $(2, 0)$, and $(1, 3)$ is shown in the following figure.
[Figure: A triangle with corners at the origin, $(2, 0)$, and $(1, 3)$.]

- (a) Find a transformation $T : S \rightarrow R, T(u, v) = (x, y) = (au + bv, cu + dv)$, where a, b, c and d are real numbers with $ad - bc \neq 0$ such that $T^{-1}(0, 0) = (0, 0)$, $T^{-1}(2, 0) = (1, 0)$, and $T^{-1}(1, 3) = (0, 1)$.
- (b) Use the transformation T to find the area $A(R)$ of the region R .

In the following exercises, use the transformation $u = y - x, v = y$, to evaluate the integrals on the parallelogram R of vertices $(0, 0)$, $(1, 0)$, $(2, 1)$, and $(1, 1)$ shown in the following figure.

[Figure: A parallelogram with corners at the origin, $(1, 0)$, $(1, 1)$, and $(2, 1)$.]

35. $\iint_R (y - x) dA$

36. $\iint_R (y^2 - xy) dA$

In the following exercises, use the transformation $y - x = u, x + y = v$ to evaluate the integrals on the square R determined by the lines $y = x, y = -x + 2, y = x + 2$, and $y = -x$ shown in the following figure.

[Figure: A square with side lengths square root of 2 rotated 45 degrees with one corner at the origin and another at $(1, 1)$.]

37. $\iint_R e^{x+y} dA$

38. $\iint_R \sin(x - y) dA$

In the following exercises, use the transformation $x = u, 5y = v$ to evaluate the integrals on the region R bounded by the ellipse $x^2 + 25y^2 = 1$ shown in the following figure.

[Figure: An ellipse with center at the origin, major axis 2, and minor 0.4.]

39. $\iint_R \sqrt{x^2 + 25y^2} dA$

40. $\iint_R (x^2 + 25y^2)^2 dA$

In the following exercises, use the transformation $u = x + y, v = x - y$ to evaluate the integrals on the trapezoidal region R determined by the points $(1, 0), (2, 0), (0, 2)$, and $(0, 1)$ shown in the following figure.

[Figure: A trapezoid with corners at $(1, 0), (0, 1), (0, 2)$, and $(2, 0)$.]

41. $\iint_R (x^2 - 2xy + y^2)e^{x+y} dA$

42. $\iint_R (x^3 + 3x^2y + 3xy^2 + y^3) dA$

43. The circular annulus sector R bounded by the circles $4x^2 + 4y^2 = 1$ and $9x^2 + 9y^2 = 64$, the line $x = y\sqrt{3}$, and the y -axis is shown in the following figure. Find a transformation T from a rectangular region S in the $r\theta$ -plane to the region R in the xy -plane. Graph S .

[Figure: In the first quadrant, a section of an annulus described by an inner radius of 0.5, outer radius slightly more than 2.5, and center the origin. There is a line dividing this annulus that comes from approximately a 30 degree angle. The portion corresponding to 60 degrees is shaded.]

44. The solid R bounded by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0, z = 1, x = 0$, and $y = 0$ is shown in the following figure. Find a transformation T from a cylindrical box S in $r\theta z$ -space to the solid R in xyz -space.

[Figure: A quarter of a cylinder with height 1 and radius 3. The center axis is the z axis.]

45. Show that $\iint_R f(\sqrt{\frac{x^2}{3} + \frac{y^2}{5}}) dA = 2\pi\sqrt{15} \int_0^1 f(\rho)\rho d\rho$, where f is a continuous function on $[0, 1]$ and R is the region bounded by the ellipse $5x^2 + 3y^2 = 15$.
46. Show that $\iiint_R f(\sqrt{16x^2 + 4y^2 + z^2}) dV = \frac{\pi}{2} \int_0^1 f(\rho)\rho^2 d\rho$, where f is a continuous function on $[0, 1]$ and R is the region bounded by the ellipsoid $16x^2 + 4y^2 + z^2 = 1$.
47. [T] Find the area of the region bounded by the curves $xy = 1$, $xy = 3$, $y = 2x$, and $y = 3x$ by using the transformation $u = xy$ and $v = \frac{y}{x}$. Use a computer algebra system (CAS) to graph the boundary curves of the region R .
48. [T] Find the area of the region bounded by the curves $xy = 2$, $xy = 3$, $y = x$, and $y = 2x$ by using the transformation $u = xy$ and $v = \frac{y}{x}$.
49. Evaluate the triple integral $\int_0^1 \int_1^2 \int_z^{z+1} (y+1) dx dy dz$ by using the transformation $u = x - z$, $v = 3y$, and $w = \frac{z}{2}$.
50. Evaluate the triple integral $\int_0^2 \int_4^6 \int_{3z}^{3z+2} (5 - 4y) dx dz dy$ by using the transformation $u = x - 3z$, $v = 4y$, and $w = z$.
51. A transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(u, v) = (x, y)$ of the form $x = au + bv$, $y = cu + dv$, where a, b, c , and d are real numbers, is called linear. Show that a linear transformation for which $ad - bc \neq 0$ maps parallelograms to parallelograms.
52. The transformation $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T_\theta(u, v) = (x, y)$, where $x = u \cos \theta - v \sin \theta$, $y = u \sin \theta + v \cos \theta$, is called a rotation of angle θ . Show that the inverse transformation of T_θ satisfies $T_\theta^{-1} = T_{-\theta}$, where $T_{-\theta}$ is the rotation of angle $-\theta$.
53. [T] Use the results of the previous exercise to find the region S in the uv -plane whose image through a rotation of angle $\frac{\pi}{4}$ is the region R enclosed by the ellipse $x^2 + 4y^2 = 1$. Use a CAS to answer the following questions.
- Graph the region S .
 - Evaluate the integral $\iint_S e^{-2uv} du dv$. Round your answer to two decimal places.
54. [T] The transformations $T_i : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $i = 1, \dots, 4$, defined by $T_1(u, v) = (u, -v)$, $T_2(u, v) = (-u, v)$, $T_3(u, v) = (-u, -v)$, and $T_4(u, v) = (v, u)$ are called reflections about the x -axis, y -axis, origin, and the line $y = x$, respectively.
- Find the image of the region $S = \{(u, v) | u^2 + v^2 - 2u - 4v + 1 \leq 0\}$ in the xy -plane through the transformation $T_1 \circ T_2 \circ T_3 \circ T_4$.
 - Use a CAS to graph the image of S .
 - Evaluate the integral $\iint_S \sin(u^2) du dv$ by using a CAS. Round your answer to two decimal places.
55. [T] The transformation $T_{k,1,1} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $T_{k,1,1}(u, v, w) = (x, y, z)$ of the form $x = ku$, $y = v$, $z = w$, where $k \neq 1$ is a positive real number, is called a stretch if $k > 1$ and a compression if $0 < k < 1$ in the x -direction. Use a CAS to evaluate the integral $\iiint_S e^{-(4x^2+9y^2+25z^2)} dx dy dz$ on the solid $S = \{(x, y, z) | 4x^2 + 9y^2 + 25z^2 \leq 1\}$ by considering the compression $T_{2,3,5}(u, v, w) = (x, y, z)$ defined by $x = \frac{u}{2}$, $y = \frac{v}{3}$, and $z = \frac{w}{5}$. Round your answer to four decimal places.

56. [T] The transformation $T_{a,0} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T_{a,0}(u, v) = (u + av, v)$, where $a \neq 0$ is a real number, is called a shear in the x -direction. The transformation, $T_{0,b} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T_{0,b}(u, v) = (u, bu + v)$, where $b \neq 0$ is a real number, is called a shear in the y -direction.
- Find transformations $T_{0,2} \circ T_{3,0}$.
 - Find the image R of the trapezoidal region S bounded by $u = 0, v = 0, v = 1$, and $v = 2 - u$ through the transformation $T_{0,2} \circ T_{3,0}$.
 - Use a CAS to graph the image R in the xy -plane.
 - Find the area of the region R by using the area of region S .
57. Use the transformation, $x = au, y = av, z = cw$ and spherical coordinates to show that the volume of a region bounded by the spheroid $\frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} = 1$ is $\frac{4\pi a^2 c}{3}$.
58. Find the volume of a football whose shape is a spheroid $\frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} = 1$ whose length from tip to tip is 11 inches and circumference at the center is 22 inches. Round your answer to two decimal places.
59. [T] Lamé ovals (or superellipses) are plane curves of equations $(\frac{x}{a})^n + (\frac{y}{b})^n = 1$, where a, b , and n are positive real numbers.
- Use a CAS to graph the regions R bounded by Lamé ovals for $a = 1, b = 2, n = 4$ and $n = 6$, respectively.
 - Find the transformations that map the region R bounded by the Lamé oval $x^4 + y^4 = 1$, also called a squircle and graphed in the following figure, into the unit disk.
[Figure: A square of side length 2 with rounded corners.]
 - Use a CAS to find an approximation of the area $A(R)$ of the region R bounded by $x^4 + y^4 = 1$. Round your answer to two decimal places.
60. [T] Lamé ovals have been consistently used by designers and architects. For instance, Gerald Robinson, a Canadian architect, has designed a parking garage in a shopping center in Peterborough, Ontario, in the shape of a superellipse of the equation $(\frac{x}{a})^n + (\frac{y}{b})^n = 1$ with $\frac{a}{b} = \frac{9}{7}$ and $n = e$. Use a CAS to find an approximation of the area of the parking garage in the case $a = 900$ yards, $b = 700$ yards, and $n = 2.72$.